

3975. Filter Occupied Intervals

Solved

Medium   Hint

You are given a 2D integer array `occupiedIntervals`, where `occupiedIntervals[i] = [starti, endi]` represents a time interval during which you are occupied. Each interval starts at `starti` and ends at `endi`, **inclusive**. These intervals may **overlap**.

You are also given two integers `freeStart` and `freeEnd`, which define a free time interval from `freeStart` to `freeEnd`, inclusive.

Your task is to merge **all** occupied intervals that overlap or touch, then remove **all** integer points in the free interval from the merged occupied intervals.

Two intervals touch if the second interval starts **immediately after** the first one ends. For example, `[1, 1]` and `[2, 2]` touch and should be merged into `[1, 2]`.

Return the **remaining** occupied intervals in **sorted** order. The returned intervals must be **non-overlapping** and must contain the **minimum** number of intervals possible. If there are no remaining occupied points, return an empty list.

Example 1:

Input: `occupiedIntervals = [[2,6],[4,8],[10,10],[10,12],[14,16]]`, `freeStart = 7`, `freeEnd = 11`

Output: `[[2,6],[12,12],[14,16]]`

Explanation:

- After merging, the occupied intervals are `[2, 8]`, `[10, 12]`, and `[14, 16]`.
- Excluding the free interval `[7, 11]` results in `[2, 6]`, `[12, 12]`, and `[14, 16]`.

Example 2:

Input: `occupiedIntervals = [[1,5],[2,3]]`, `freeStart = 3`, `freeEnd = 8`

Output: `[[1,2]]`

Explanation:

- After merging, the occupied interval is `[1, 5]`.
- Excluding the free interval `[3, 8]` results in `[1, 2]`.

Constraints:

- `1 <= occupiedIntervals.length <= 5 * 104`
- `occupiedIntervals[i].length == 2`
- `1 <= starti <= endi <= 109`
- `1 <= freeStart <= freeEnd <= 109`

Seen this question in a real interview before? 1/6

Yes No

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